Soft Hair on PDL Closures: A Structural Correspondence Between Surface Microstate Counting and Black Hole Hair

PDL Document D64 — Two New Open Problems (OP-D64-1, OP-D64-2)

Cédric Laubscher

Independent Researcher, Switzerland

cedriclaubscher.ch

June 2026

Abstract

Document D50 proves, as an unconditional theorem of the four PDL axioms C1–C4, that the active surface of a PDL proton closure carries $\Omega_{\text{surf}} = 4^{R_{\text{surf}}}$ statistically independent microstates at fixed global quantum numbers (the proton quintuplet), while the interior carries none ($\Omega_{\text{val}} = 1$). This document establishes, for the first time explicitly, the structural correspondence between this result and the soft-hair programme of Hawking, Perry and Strominger [1, 2], in which black holes carry an infinite-dimensional space of supertranslation and superrotation charges at fixed mass, charge and angular momentum. Both results share the same logical architecture: a macroscopic object fully characterised by a small number of global invariants, yet carrying a combinatorially large space of boundary degrees of freedom invisible to any measurement of the global invariants alone. The correspondence is presented as a structural analogy with independent motivation on both sides, not as a claim of mathematical identity, following the cross-framework identification protocol already in use for the PDL–OFN bridge [3]. The document also reports a negative result: a naive extrapolation of the PDL per-relation degeneracy to macroscopic black holes (combinatorial pairing of engaged nucleons) fails by 36.7 orders of magnitude against the Bekenstein–Hawking value, while a surface-restricted count narrows this to 1.4 orders of magnitude, indicating that the relevant microscopic degrees of freedom are areal, not volumetric, in nature. Two open problems are formally introduced: **OP-D64-1**, the derivation of the macroscopic many-nucleon entropy law from PDL combinatorics alone (the genuine microstate-counting problem, equivalent in difficulty to OP1 of D38), and **OP-D64-2**, the construction of a fully relational unit system in which the speed of light c and the proton–electron mass ratio μ^* are derived from C1–C4 rather than imported from measurement, a necessary precondition for an autonomous PDL derivation of $\lambda_{\text{PDL}} = 4\ell_{\text{Pl}}^2$ (the principal open problem of D37).

Contents

1 Purpose and intended audience

This document is written to be read by colleagues outside the PDL programme who are familiar with black hole thermodynamics and the soft-hair literature, but not with the internal combinatorics of PDL. Section 2 therefore recalls, without proof, exactly the PDL results required; the proofs themselves are in D29, D37, D38, D42 and D50, cited throughout. Section 3 recalls the relevant external literature in the same spirit, for readers approaching from the PDL side. Section 4 states the correspondence precisely, with explicit care about what is and is not being claimed. Section 5 reports a negative numerical result that delimits the correspondence. Section 6 introduces the two new open problems. No result in this document modifies or supersedes D50, D37, or D38; it is a correspondence document, not a derivation document.

2 PDL prerequisites: the surface degeneracy theorem of D50

The PDL programme derives fundamental physical structures from four combinatorial axioms (C1–C4) on finite signed graphs, with a single irreducible external parameter, $\Delta m_{\rm iso} = m_d - m_u$, at the PDL–QCD interface [4]. The proton is identified as the unique hierarchical composite satisfying these axioms, characterised by the integer quintuplet $(n_u, n_d, r_{\rm val}, R_{\rm sea}, R_{\rm tot}) = (24, 28, 930, 10\,087, 11\,017)$ [5]. The electron is identified with the minimal closure K_4, the complete graph on four vertices and six edges, the smallest signed graph satisfying C1–C4 [4].

Definition 2.1 (Active surface, [6, 7]). *The active surface $R_{\rm surf} = 310\varphi \in \mathbb{Q}(\sqrt{5})$, where $\varphi = (1+\sqrt{5})/2$, is the subset of the proton's total relational budget $R_{\rm tot}$ that forms stable $(A)\wedge(B)$ couplings with an external K_4 closure (the electron prototype). This is an unconditional theorem of C1–C4.*

Definition 2.2 (Mixed triangles and the $(A)\wedge(B)$ stability criterion, [8]). *A mixed triangle is a triangle of the PDL relational graph formed by two vertices e_i, e_j of the external K_4 closure and one relation p_k of the proton. PDL structures pulsate through two alternating half-cycles, $c \in \{1, 2\}$ (the discrete analogue of a phase). For a fixed edge (e_i, e_j) of K_4, with intrinsic sign $s^{(1)}(e_i, e_j) \in \{+1, -1\}$ fixed by the coherent configuration of K_4, a* cross-sign configuration *for p_k is a choice of four binary cross-edge signs $(s_{i,1}, s_{j,1}, s_{i,2}, s_{j,2}) \in \{+1, -1\}^4$, one for each of the two K_4 vertices at each of the two half-cycles, giving $2^4 = 16$ possibilities in total. Define the cross-products $P_c = s_{i,c} \cdot s_{j,c}$ for $c = 1, 2$. The cross-sign configuration is $(A)\wedge(B)$*-stable *if and only if*

$$(A) \quad P_1 = s^{(1)}(e_i, e_j), \qquad (B) \quad P_2 = -P_1. \tag{1}$$

Condition (A) requires the relation to couple in phase with K_4 at the first half-cycle; condition (B) requires it to couple in antiphase at the second, the discrete signature of a complete pulsation cycle. Exhaustive enumeration over all $8 \times 6 \times 16 = 768$ combinations of (K_4 configuration, edge, cross-sign choice) shows that exactly 4 of the 16 cross-sign configurations satisfy (A)∧(B) for any fixed edge sign, independently of which of the 8 coherent configurations of K_4 is realised [8, 9].

Theorem 2.3 (Surface locality of entropy, BH-1, [10]). *The valence sector of the proton admits exactly one externally distinguishable $(A)\wedge(B)$-stable configuration: $\Omega_{\rm val} = 1$, hence $S_{\rm val} = 0$. The sea sector is excluded from stationary coupling, hence $S_{\rm sea} = 0$. All entropy is localised on the active surface.*

Theorem 2.4 (Surface degeneracy, D50 [9]). *Each surface relation $p_k \in R_{\mathrm{surf}}$ admits exactly 4 statistically independent $(A) \wedge (B)$-stable cross-sign configurations out of 16 possible (Definition 2.2), a fraction proved by exhaustive enumeration over the 768 configurations of D29 [8]. Independence across distinct surface relations is proved via the three lemmas of D42 (equiparticipation, cross-triangle blindness, S_4-equivariance verified over 24 576 cases) and confirmed by direct enumeration over 30 720 cross-sign pairs: $P(e_1 \wedge e_2\,\mathrm{stable}) = 1/16 = (1/4)^2$ exactly. Consequently*

$$\Omega_{\mathrm{surf}} = 4^{R_{\mathrm{surf}}}, \qquad S_{\mathrm{surf}} = k_B R_{\mathrm{surf}} \ln 4 = 695.352\ \mathrm{nats}. \tag{2}$$

This is an unconditional theorem of C1–C4; no free parameter enters.

The physical reading of Theorem 2.4, given informally in D50, is that each of the $R_{\mathrm{surf}} \approx 501.6$ surface relations behaves as a two-bit switch: of the 16 possible ways an external K_4 can couple to it, exactly 4 are stable, and which of these 4 is realised is, as far as any external probe restricted to the global quintuplet is concerned, unobservable. The proton's global quantum numbers (its quintuplet) do not determine which of the $4^{R_{\mathrm{surf}}}$ stable surface configurations is realised.

3 External prerequisites: the no-hair theorem and soft hair

The classical no-hair theorem [11, 12] states that a stationary black hole in General Relativity is fully characterised, up to diffeomorphism, by three parameters: mass M, charge Q, and angular momentum J. Any additional information about the matter that formed the black hole appears, classically, to be lost — the seed of the black hole information paradox [13, 14].

Hawking, Perry and Strominger [1] showed that this picture is incomplete. Black holes admit an infinite-dimensional family of soft (zero-energy) supertranslation charges defined on the horizon, associated with the Bondi–van der Burg–Metzner–Sachs (BMS) asymptotic symmetry group. Two black holes with identical M, Q, J can carry different supertranslation charges — soft hair — which are in principle measurable at the horizon even though they are invisible to any measurement of M, Q, J alone. The follow-up work [2] extends this to superrotation charges associated with horizon supertranslation and superrotation hair, and the broader literature [15, 16] situates this within the infrared structure of gravity. The physical upshot, as stated in [1], is that complete information about the soft quantum state of a black hole is stored holographically on the horizon, even though it is invisible to the macroscopic, no-hair description.

4 The structural correspondence

Both results above share an identical logical skeleton, summarised in Table 1.

Table 1: The structural correspondence between black hole soft hair and PDL surface microstates.

Black hole (GR, [1, 2])	PDL proton closure (D29, D37, D42, D50)
Global invariants: M, Q, J	Global invariants: the quintuplet $(n_u, n_d, r_{\text{val}}, R_{\text{sea}}, R_{\text{tot}})$
No-hair theorem: M, Q, J fix the geometry up to diffeomorphism	$\Omega_{\text{val}} = 1$: the interior has no externally distinguishable degeneracy (Thm. 2.3)
Soft supertranslation charges on the horizon, invisible to M, Q, J	Surface relations $p_k \in R_{\text{surf}}$, with 4 stable cross-sign realisations each, invisible to the quintuplet
Information stored holographically on the horizon [1]	Entropy localised entirely on R_{surf}, never on the interior (Thm. 2.3)
Degeneracy at fixed global charges: continuous (BMS is infinite-dimensional)	Degeneracy at fixed global charges: discrete, $\Omega_{\text{surf}} = 4^{R_{\text{surf}}}$ (Thm. 2.4)

Structural correspondence (not a proof of identity)

What is claimed. The two frameworks exhibit the same structural pattern — macroscopic global invariants insufficient to fix the microstate, with the missing information located strictly on a boundary — and each side has independent motivation for this pattern within its own framework: the no-hair theorem and BMS symmetry on the GR side; the $(A) \wedge (B)$ criterion and D42's independence lemmas on the PDL side. This satisfies the cross-framework identification protocol already applied to the PDL–OFN bridge [3]: an analogy may be asserted once each side is independently motivated, but no further identification (e.g. a literal correspondence between R_{surf} relations and BMS supertranslation modes) is claimed here.

What is not claimed. (1) That the PDL surface relations *are* BMS supertranslation modes, or that there exists a bijection between them. No such map is constructed in this document. (2) That Theorem 2.4 resolves any open problem of black hole information loss; PDL has, at this stage, no dynamical (time-dependent) theory of horizon evolution comparable to Hawking radiation [13], only the static entropy count of Theorem 2.4. (3) That the R_{surf}-scale degeneracy of a single proton extrapolates to the macroscopic degeneracy of an astrophysical black hole; this is addressed, and shown to fail under the simplest extrapolation, in Section 5.

5 A documented negative result: macroscopic extrapolation fails

Theorem 2.4 establishes $\Omega_{\mathrm{surf}} = 4^{R_{\mathrm{surf}}}$ for a single proton, with $R_{\mathrm{surf}} \approx 501.6$, giving $S_{\mathrm{surf}} \approx 695$ nats. An astrophysical black hole of $N \sim 10^{57}$ nucleons (one solar mass) requires, by the Bekenstein–Hawking formula combined with the PDL effective coupling [17], an entropy of order $\ln \Omega_{\mathrm{surf}}(N) \approx 1.06 \times 10^{77}$ nats. Whether the per-proton mechanism of Theorem 2.4 extends to this macroscopic regime is precisely D38's Open Problem 1 (the derivation of $\ln \Omega_{\mathrm{surf}}(N) = 4\pi(M_{\mathrm{eff}}/M_{\mathrm{Pl}})^2$ from PDL axioms without invoking Schwarzschild geometry), restated here as OP-D64-1 in Section 6.

Proposition 5.1 (Naive volumetric extrapolation, negative result). *Let $k = \sigma(N)N$ be the number of gravitationally engaged nucleons in an ensemble of N nucleons, under the Gate 3 engagement fraction [18]. If each pair of engaged nucleons contributes one independent bit of entropy (the direct extrapolation of the per-relation mechanism of Theorem 2.4 to pairs of whole nucleons), then for $N = 1.196 \times 10^{57}$ (one solar mass),*

$$\ln \Omega_{\mathrm{surf}}^{\mathrm{naive}} = \binom{k}{2} \ln 2 \approx 4.96 \times 10^{113} \text{ nats}, \tag{3}$$

exceeding the required value by a factor of 4.67×10^{36} (36.7 orders of magnitude). The volumetric pairwise mechanism is therefore excluded as a model of macroscopic black hole entropy.

Proposition 5.2 (Surface-restricted extrapolation, partial improvement). *Restricting the pairwise count of Proposition 5.1 to only those nucleons geometrically located at the surface of a uniform-density sphere of N nucleons ($N_{\mathrm{surf}} \sim N^{2/3}$) gives*

$$\ln \Omega_{\mathrm{surf}}^{\mathrm{surf}} \approx 4.40 \times 10^{75} \text{ nats}, \tag{4}$$

a factor of 0.041 relative to the target, i.e. an improvement from 36.7 to 1.4 orders of magnitude. This indicates that the relevant microscopic degrees of freedom for macroscopic black hole entropy are areal, consistent with the holographic principle [19, 20] and with the surface-locality result of Theorem 2.3 itself, but the remaining 1.4-order discrepancy shows that "nucleon" is not the correct counting unit; Section 6 (OP-D64-1) identifies the correct unit as the open problem.

Remark 5.3. *Both propositions are reported here as negative or partial results, in keeping with the PDL programme's convention of documenting failed approaches with the same rigour as positive ones [21]. Proposition 5.1 rules out one natural hypothesis; Proposition 5.2 narrows the search but does not resolve it.*

6 Two new open problems

Open Problem 1 (OP-D64-1 — Macroscopic many-nucleon surface counting). *Derive $\ln \Omega_{\mathrm{surf}}(N) = 4\pi(M_{\mathrm{eff}}/M_{\mathrm{Pl}})^2$ for $N \gg 1$ nucleons directly from PDL combinatorics, without invoking the Schwarzschild relation $A = 16\pi(GM/c^2)^2$ as an external input (cf. D38's Open Problem 1 [17]). Proposition 5.1 excludes the naive pairwise-volumetric mechanism; Proposition 5.2 indicates that the correct counting unit is areal rather than nucleonic, narrowing but not resolving the search. This problem is, in its full generality, equivalent to the long-standing problem of microscopic black hole entropy counting addressed for special (extremal) cases by other frameworks [22]; no claim is made that PDL resolves it here.* ***Priority:*** *high.* ***Entry point:*** *D38, this document.*

Open Problem 2 (OP-D64-2 — The spacetime bridge: relational c and μ^*)**.** *The reduced Planck constant $\hbar$ is already interpreted, without reference to spacetime, as the action quantum of one K_4 pulsation cycle [4], and Newton's constant G is an unconditional PDL theorem [23, 24, 25]. However, the speed of light c enters the PDL corpus only as a qualitative reinterpretation of the standard limiting speed of information transfer [4], and the proton–electron mass ratio $\mu^* = m_p/m_e$ remains a strong conjecture with an unresolved 47 ppm residual (OP7 [26, 27]), reclassified in DS01 [28] as a metrological interface problem consistent with known QED isospin-breaking corrections rather than a structural gap. Resolving the principal open problem of D37 — the prediction $\lambda_{\mathrm{PDL}} = 4\ell_{\mathrm{Pl}}^2$, requiring a counting argument that each Planck area corresponds to exactly four relational units of R_{surf} [10] — requires expressing $l_{\mathrm{Pl}}^2 = \hbar G/c^3$ entirely in relational units. This requires, in turn: (a) a combinatorial derivation of c as a propagation rate intrinsic to the PDL relational network, with no antecedent reference to distance or duration in the standard sense; and (b) the promotion of μ^* from conjecture to theorem. Neither exists in the corpus at this stage.* ***Priority:*** *high.* ***Entry point:*** *D01, D28, D30, D37, DS01, this document.*

7 Epistemic status

Table 2: Epistemic status of the results of D64.

Result	Status	Dependency
$\Omega_{\mathrm{surf}} = 4^{R_{\mathrm{surf}}}$ (recalled, not re-derived)	Theorem	D29, D42, D50
Structural correspondence with soft hair (Table 1)	Structural analogy, independently motivated	this document, [1, 2, 3]
Naive pairwise extrapolation excluded (Prop. 5.1)	Negative result, verified by direct computation	this document
Surface-restricted extrapolation (Prop. 5.2)	Partial improvement, not a resolution	this document
OP-D64-1 (macroscopic counting)	Open	D38, this document
OP-D64-2 (relational c, μ^*)	Open	D01, D28, D30, D37, DS01, this document

8 Summary for colleagues outside the PDL programme

A reader unfamiliar with PDL may retain the following. PDL is a combinatorial framework that derives physical structures from finite signed graphs satisfying four axioms, with the proton modelled as a specific such graph. Within this framework, it has been proved (Theorem 2.4, document D50) that the boundary of this graph carries an enormous number of microscopic configurations — $4^{R_{\mathrm{surf}}}$, with $R_{\mathrm{surf}} \approx 502$ — that are indistinguishable from one another by any measurement of the proton's global quantum numbers, while the interior carries none. This is, independently of any intended connection to gravitational physics, structurally the same statement as the soft-hair result of Hawking, Perry and Strominger [1, 2]: black holes carry boundary degrees of freedom invisible to their mass, charge and angular momentum. This document states that correspondence explicitly for the first time, without claiming any further identification between the two frameworks, and uses it to motivate two precise open problems: whether the PDL mechanism scales correctly to real, macroscopic black holes (OP-D64-1), and whether PDL can express its own fundamental constants without borrowing c and m_p/m_e from measurement (OP-D64-2).

References

[1] S. W. Hawking, M. J. Perry, and A. Strominger. Soft hair on black holes. *Physical Review Letters*, 116(23):231301, 2016. https://doi.org/10.1103/PhysRevLett.116.231301, arXiv:1601.00921.

[2] S. W. Hawking, M. J. Perry, and A. Strominger. Superrotation charge and supertranslation hair on black holes. *Journal of High Energy Physics*, 2017(5):161, 2017. https://doi.org/10.1007/JHEP05(2017)161, arXiv:1611.09175.

[3] C. Laubscher and O. I. Evdokimov. From $\mathbb{Z}_3$ to three generations: A pdl–ofn bridge (n02), 2026. Unpublished manuscript, pending co-author review; not present in the Zenodo DOI registry at time of citation.

[4] C. Laubscher. The emergence of physical reality within the framework of the projective dynamic logo (pdl). *Zenodo*, 2026. D01, https://doi.org/10.5281/zenodo.18462686.

[5] C. Laubscher. On the combinatorial selection of the proton architecture in pdl: Axiomatic constraints, integer structures, and a uniqueness conjecture. *Zenodo*, 2026. D16, https://doi.org/10.5281/zenodo.18832953.

[6] C. Laubscher. A minimal relational sketch for the emergence of the golden ratio in pdl. *Zenodo*, 2026. D05, https://doi.org/10.5281/zenodo.18581453.

[7] C. Laubscher. Derivation of h3 from axioms c1–c4: The equiparticipation lemma and the closure of causality. *Zenodo*, 2026. D42, https://doi.org/10.5281/zenodo.20041348.

[8] C. Laubscher. Axiomatic derivation of the leading valence engagement amplitude in the pdl framework: Gate 1 resolved. *Zenodo*, 2026. D29, https://doi.org/10.5281/zenodo.19283107.

[9] C. Laubscher. Derivation of the bekenstein–hawking one-quarter coefficient from the pdl axioms: Resolution of op12/bh-3. *Zenodo*, 2026. D50, https://doi.org/10.5281/zenodo.20029777.

[10] C. Laubscher. Surface locality of relational entropy and the pdl area law. *Zenodo*, 2026. D37, https://doi.org/10.5281/zenodo.19354096.

[11] W. Israel. Event horizons in static vacuum space-times. *Physical Review*, 164:1776–1779, 1967. https://doi.org/10.1103/PhysRev.164.1776.

[12] B. Carter. Axisymmetric black hole has only two degrees of freedom. *Physical Review Letters*, 26:331–333, 1971. https://doi.org/10.1103/PhysRevLett.26.331.

[13] S. W. Hawking. Particle creation by black holes. *Communications in Mathematical Physics*, 43:199–220, 1975. https://doi.org/10.1007/BF02345020.

[14] S. W. Hawking. Breakdown of predictability in gravitational collapse. *Physical Review D*, 14:2460–2473, 1976. https://doi.org/10.1103/PhysRevD.14.2460.

[15] A. Strominger. On bms invariance of gravitational scattering. *Journal of High Energy Physics*, 2014(7):152, 2014. https://doi.org/10.1007/JHEP07(2014)152, arXiv:1312.2229.

[16] L. Donnay, G. Giribet, H. A. González, and M. Pino. Supertranslations and superrotations at the black hole horizon. *Physical Review Letters*, 116(9):091101, 2016. https://doi.org/10.1103/PhysRevLett.116.091101.

[17] C. Laubscher. Bekenstein–hawking entropy from the pdl effective gravitational coupling: the 4π identity, surface counting, and primordial black hole predictions. *Zenodo*, 2026. D38, https://doi.org/10.5281/zenodo.19354682.

[18] C. Laubscher. Proof of $g_{\mathrm{eff}}(n) = \sigma(n) \cdot g_{\mathrm{pdl}}$ from the trace structure of k_4 and independence of gravitational engagement. *Zenodo*, 2026. D36, https://doi.org/10.5281/zenodo.19323033.

[19] G. 't Hooft. Dimensional reduction in quantum gravity. *arXiv preprint gr-qc/9310026*, 1993. https://arxiv.org/abs/gr-qc/9310026.

[20] L. Susskind. The world as a hologram. *Journal of Mathematical Physics*, 36:6377–6396, 1995. https://doi.org/10.1063/1.531249.

[21] C. Laubscher. Pdl global mapping of structures, results, and open problems (version 29). *Zenodo*, 2026. DM v29, https://doi.org/10.5281/zenodo.20701571.

[22] A. Strominger and C. Vafa. Microscopic origin of the bekenstein-hawking entropy. *Physics Letters B*, 379:99–104, 1996. https://doi.org/10.1016/0370-2693(96)00345-0.

[23] C. Laubscher. Universal coherence leakage: A structural bridge between the gravitational and electromagnetic constants in the projective dynamic logo framework (version 3). *Zenodo*, 2026. D21, https://doi.org/10.5281/zenodo.19056994.

[24] C. Laubscher. The causal chain of physical reality: From four axioms to newton's constant via the geometric leakage parameter. *Zenodo*, 2026. D43, https://doi.org/10.5281/zenodo.19678389.

[25] C. Laubscher. Closure of op-b: Derivation of the hierarchical filter factor k from the pdl axioms. *Zenodo*, 2026. D44, https://doi.org/10.5281/zenodo.19678474.

[26] C. Laubscher. The pdl–qcd boundary: Structural derivation of the proton–electron mass ratio correction and a parameter-free conjecture for newton's constant. *Zenodo*, 2026. D28, https://doi.org/10.5281/zenodo.19282932.

[27] C. Laubscher. Structural derivation of the qcd correction coefficient in the pdl mass ratio formula and completion of the ε_g conjecture: Gate 2 resolved. *Zenodo*, 2026. D30, https://doi.org/10.5281/zenodo.19294449.

[28] C. Laubscher. Projective dynamic logo: Programme closure at d55 – a provisional synthesis. *Zenodo*, 2026. DS01, https://doi.org/10.5281/zenodo.20187274.